

Recap:

Linear Systems. $X' = AX$

Linearity Principle. $kX_1, X_1 + X_2$

General Solution: $X(t) = c_1 \cdot X_1 + c_2 X_2$

Matrix Exponential $X' = AX, X(0) = X_0$ $X(t) = e^{At} X_0 = P \cdot e^{Jt} P^{-1} X_0$

I. $\lambda_1 \neq \lambda_2$ real eigenvalues.

$$e^{At} X_0 = P \cdot e^{Jt} \cdot \overset{P^{-1}X_0}{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}} = \underbrace{c_1 v_1 e^{\lambda_1 t}}_{X_1} + \underbrace{c_2 v_2 e^{\lambda_2 t}}_{X_2}$$

II. $\lambda \neq \bar{\lambda}$ complex eigenvalues

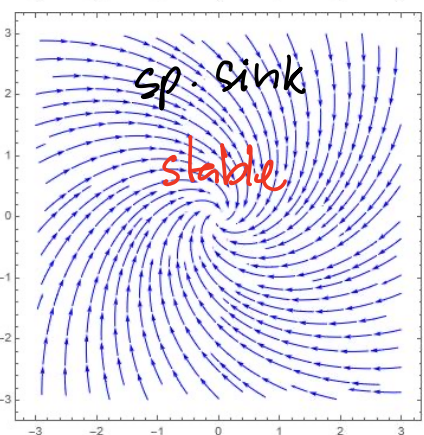
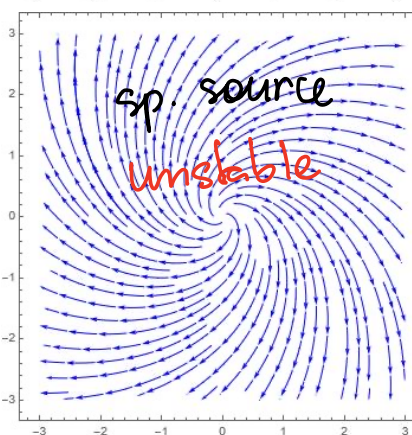
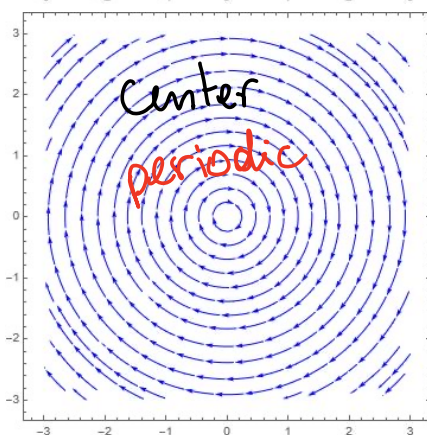
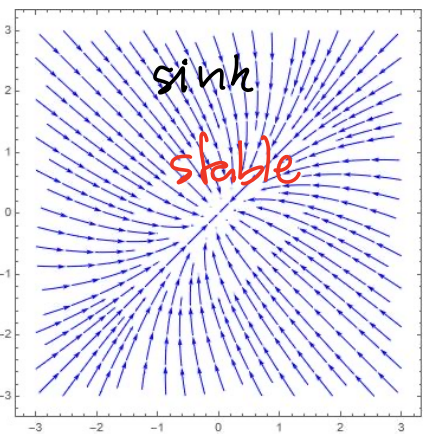
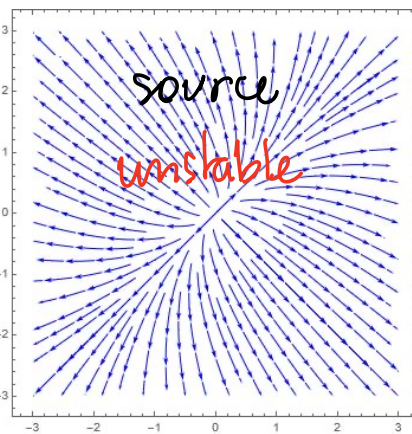
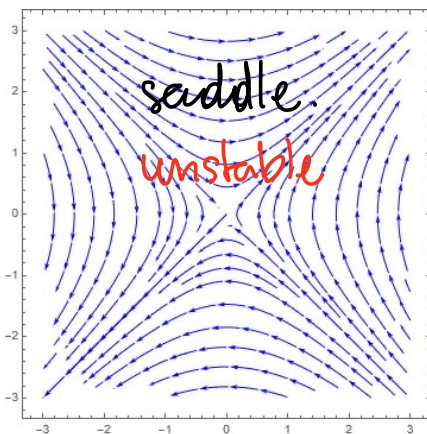
$$e^{At} x_0 = P \cdot e^{Jt} \cdot P^{-1} x_0 = \underbrace{P \cdot e^{Jt}}_{\text{complex}} \cdot \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$= c_1 \cdot \underbrace{v \cdot e^{\lambda t}}_{\tilde{x}_1} + c_2 \cdot \underbrace{\bar{v} \cdot e^{\bar{\lambda} t}}_{\tilde{x}_2}$$

$$x_1 = \frac{\tilde{x}_1 + \tilde{x}_2}{2} = \operatorname{Re}(v e^{\lambda t}), \quad x_2 = \frac{\tilde{x}_1 - \tilde{x}_2}{2} = \operatorname{Im}(v e^{\lambda t})$$

$$x(t) = c_1 \cdot \operatorname{Re}(v e^{\lambda t}) + c_2 \cdot \operatorname{Im}(v e^{\lambda t})$$

Types of equilibrium points:



Lecture 09

3.5 Repeated Eigenvalues and special cases.

"3.6" 2nd order as systems

3.7. Trace-Determinant Plane

3.5. Repeated Eigenvalues.

Consider $X' = AX$ with rep. eig. $\lambda \neq 0$.

a) if geometric multiplicity is 2:

$$e^{At} = P e^{Jt} P^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} e^{\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} t} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^{\lambda t} & 0 \\ 0 & e^{\lambda t} \end{bmatrix}$$

$$e^{At} = e^{\lambda I t} = e^{\lambda t} \cdot I$$

b) if geometric multiplicity is 1.

$$e^{At} = P e^{Jt} P^{-1} = P e^{\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t} P^{-1}$$

Computing $e^{\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t}$:

$$t^2 J^2 = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t^2 = \begin{bmatrix} \lambda^2 t^2 & 2\lambda t^2 \\ 0 & \lambda^2 t^2 \end{bmatrix}$$

$$t^3 J^3 = \begin{bmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{bmatrix} \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t^3 = \begin{bmatrix} \lambda^3 t^3 & 3\lambda^2 t^3 \\ 0 & \lambda^3 t^3 \end{bmatrix} \quad (tA)^n = \begin{bmatrix} (\lambda t)^n & nt(\lambda t)^{n-1} \\ 0 & (\lambda t)^n \end{bmatrix}$$

$$t^4 J^4 = \begin{bmatrix} \lambda^3 & 3\lambda^2 \\ 0 & \lambda^3 \end{bmatrix} \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t^4 = \begin{bmatrix} \lambda^4 t^4 & 4\lambda^3 t^4 \\ 0 & \lambda^4 t^4 \end{bmatrix}$$

$$\begin{aligned} e^{Jt} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} \lambda t & t \\ 0 & \lambda t \end{bmatrix} + \frac{1}{2!} \begin{bmatrix} (\lambda t)^2 & 2\lambda t^2 \\ 0 & (\lambda t)^2 \end{bmatrix} + \frac{(\lambda t)^3}{3!} \begin{bmatrix} (\lambda t)^3 & 3\lambda^2 t^3 \\ 0 & (\lambda t)^3 \end{bmatrix} + \dots \\ &= \begin{bmatrix} 1 + \lambda t + \frac{(\lambda t)^2}{2!} + \frac{(\lambda t)^3}{3!} + \dots & \left(0 + t + \frac{2\lambda t^2}{2! \cdot 1!} + \frac{3\lambda^2 t^3}{3! \cdot 2!} + \dots + \frac{n\lambda^{n-1} t^n}{n! \cdot (n-1)!} + \dots \right) \\ 0 & 1 + \lambda t + \frac{(\lambda t)^2}{2!} + \frac{(\lambda t)^3}{3!} + \dots \end{bmatrix} = t \begin{bmatrix} \dots & \dots \\ \dots & \dots \end{bmatrix} \\ &= \begin{bmatrix} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} & t \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} \\ 0 & \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} \end{bmatrix} = \begin{bmatrix} e^{\lambda t} & t e^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix} \end{aligned}$$

Conclusion:

$$e^{\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t} = \begin{bmatrix} e^{\lambda t} & t e^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix}$$

Jordan Canonical Form.

Example: Verify that $e^{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} t} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is a solution of

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x' = x + y \\ y' = y \end{array} \quad \begin{array}{l} x(0) = 1 \\ y(0) = 1 \end{array}$$

$$\lambda = 1. \quad A - \lambda I$$

$$\text{eig. } \begin{bmatrix} 0 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \quad y=0, v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\text{gen. eig. } \begin{bmatrix} 0 & 1 & | & 1 \\ 0 & 0 & | & 0 \end{bmatrix} \Rightarrow y=1 \Rightarrow v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I = P^{-1}$$

$$X(t) = e^{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} e^t & t e^t \\ 0 & e^t \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} e^t + t e^t \\ e^t \end{bmatrix}$$

$$= \begin{bmatrix} e^t \\ 0 \end{bmatrix} + \begin{bmatrix} t e^t \\ e^t \end{bmatrix} = e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + t e^t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X' = \begin{bmatrix} (e^t) + (e^t + t e^t) \\ e^t \end{bmatrix} = \begin{bmatrix} y + x \\ y \end{bmatrix} = AX$$

In general,

$$e^{At} x_0 = c_1 \underbrace{v \cdot e^{\lambda t}}_{x_1} + c_2 \underbrace{w t e^{\lambda t}}_{x_2}$$

Ex. What is the general solution of

$$x' = 4x + y$$

$$y' = -x + 2y$$

Step 0: eigenvalues. :

a) $\lambda = 2$

~~b) $\lambda = 3$~~

c) $\lambda = -3$

d) $\lambda = -2$

pol 1.

eigenvector : $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

gen. eigenvector : $v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Step 1: Compute $P e^{Jt} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} e^{\lambda t} & t e^{\lambda t} \\ & e^{\lambda t} \end{bmatrix}$

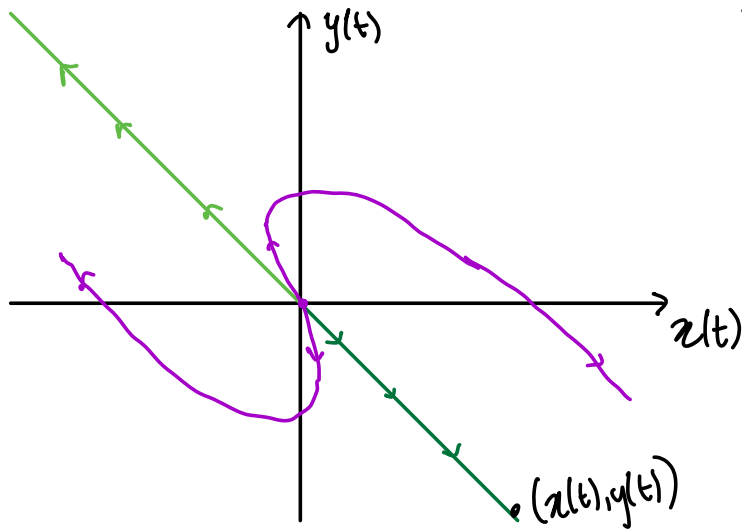
$$= \begin{bmatrix} e^{\lambda t} & t e^{\lambda t} + e^{\lambda t} \\ -e^{\lambda t} & -t e^{\lambda t} \end{bmatrix}$$

Step 2: Use $P^{-1} x_0 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ and state your solution

$$x(t) = \begin{bmatrix} c_1 e^{\lambda t} + c_2 t e^{\lambda t} + c_2 e^{\lambda t} \\ -c_1 e^{\lambda t} - c_2 t e^{\lambda t} \end{bmatrix} = c_1 \begin{bmatrix} e^{\lambda t} \\ -e^{\lambda t} \end{bmatrix} + c_2 \begin{bmatrix} t e^{\lambda t} + e^{\lambda t} \\ -t e^{\lambda t} \end{bmatrix}$$

Extra step. Sketch your solution

$$X(t) = c_1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \begin{bmatrix} te^t + e^t \\ -te^t \end{bmatrix}$$



$$-\begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t$$

$$c_1 = 1, c_2 = 0$$

$$c_1 = -1, c_2 = 0$$

$$c_1 = 0, c_2 = 1$$

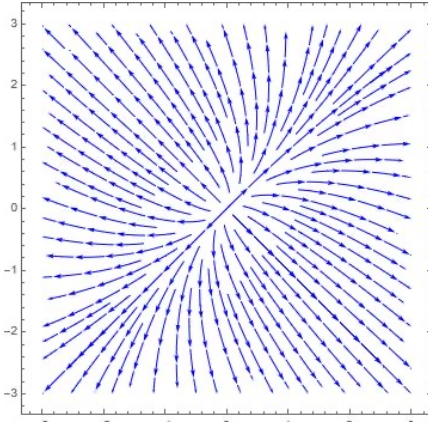
$$c_1 = 0, c_2 = -1$$

Ex : Sketch the phase portrait of $\begin{cases} x' = x \\ y' = y \end{cases}$

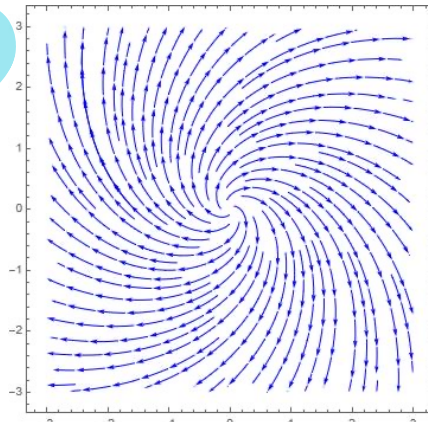
$$x = e^t \cdot x_0$$

$$y = e^t \cdot y_0$$

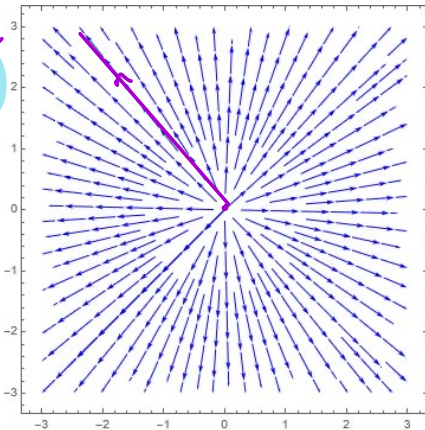
a)



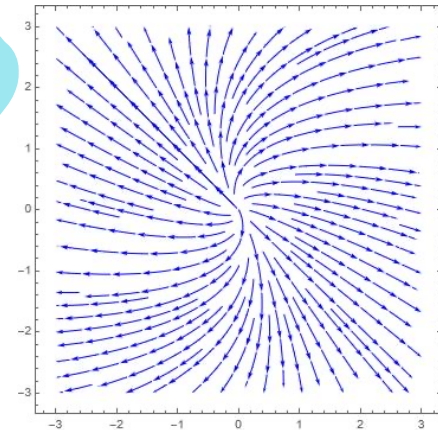
b)



~~c)~~



d)



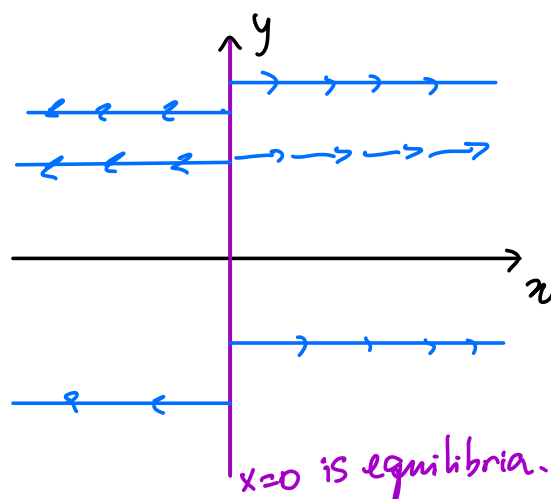
Solution: c.

Systems with zero eigenvalues.

Ⓘ Only one zero eigenvalue: $J = \begin{bmatrix} \lambda & 0 \\ 0 & 0 \end{bmatrix}$

$$\begin{cases} x' = \lambda x \Rightarrow x(t) = x_0 e^{\lambda t} \\ y' = 0 \Rightarrow y = \text{const.} \end{cases}$$

Equilibria: $(0,0)$ $(0,1)$?
 $(0,k)$?



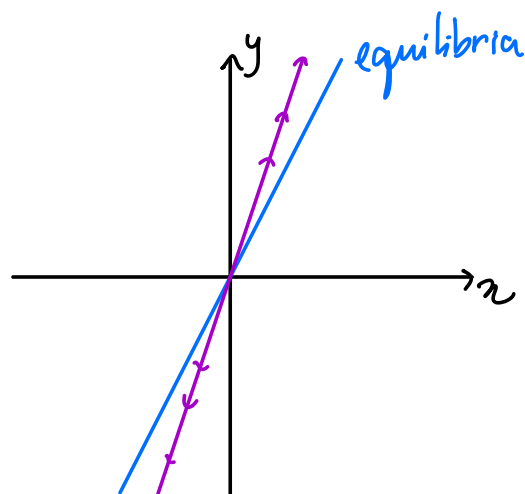
HW3
Q1

$$\begin{cases} x' = 6x - 3y \\ y' = 2x - y \end{cases}$$

$$\lambda = 5, 0.$$

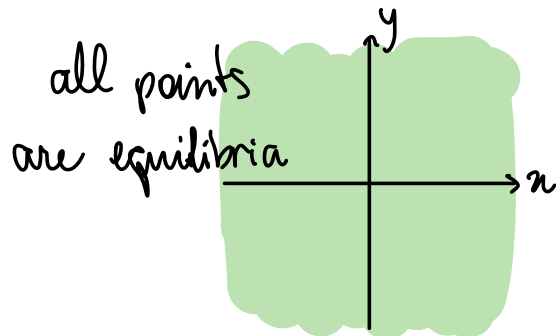
$$v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$y = 2x \Rightarrow \text{equilibria } (x, 2x)$$



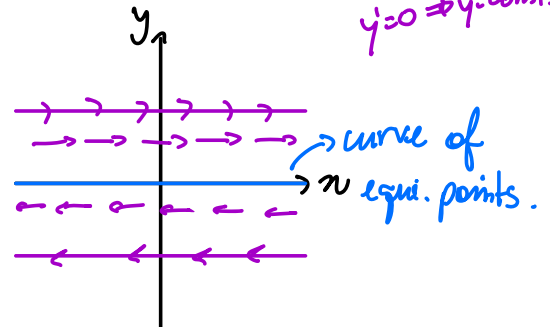
(II) if "both" eigenvalues are zero,

$$J = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{cases} x' = 0 \\ y' = 0 \end{cases}$$



$$J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{cases} x' = y = 0 \\ y' = 0 \end{cases}$$

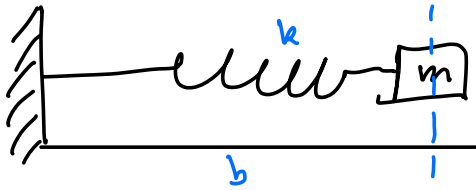
if $y > 0, x' > 0$
 $y' = 0 \Rightarrow y = \text{const.}$



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break.

3.6. Review of Harmonic Oscillator.



$$m\ddot{x} + b\dot{x} + kx = 0$$

a) Introduce $y = \dot{x}$. Write as a system

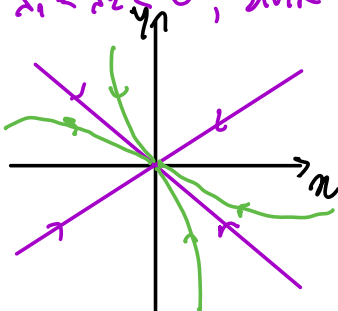
$$\begin{cases} \dot{x} = y \\ \dot{y} = -\frac{k}{m}x - \frac{b}{m}y \end{cases}$$

b) $\dot{X} = AX$. $A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}$, eigenvalues?

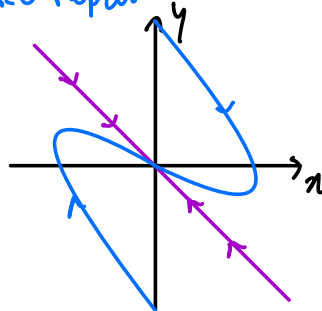
$$\det \begin{vmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\lambda - \frac{b}{m} \end{vmatrix} = \lambda^2 + \frac{b}{m}\lambda + \frac{k}{m} = 0. \Rightarrow \lambda = -\frac{b}{2m} \pm \frac{\sqrt{b^2 - 4km}}{2m}$$

c) Sketch of phase portraits

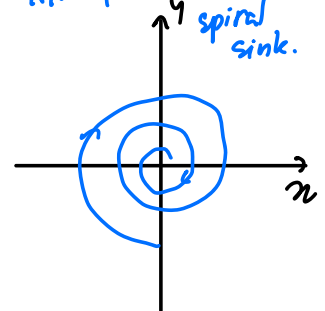
(I) ^{over} $b^2 - 4km > 0$, λ_1, λ_2
 $\lambda_1 < \lambda_2 < 0$, sink



(II) ^{crit.} $b^2 - 4km = 0$
 $\lambda < 0$ repeated root.



(III) ^{under.} $b^2 - 4km < 0$
 $\lambda, \bar{\lambda}$, $\text{Re}(\lambda) < 0$
spiral sink.



3.7. Trace - Determinant Plane.

Motivation: For the harmonic oscillator, we didn't compute the precise eigenvalues, only

$$\det \left(\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} - \begin{bmatrix} \lambda & \\ & \lambda \end{bmatrix} \right)$$

For any matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & \\ & \lambda \end{bmatrix} \right) = (a-\lambda)(d-\lambda) - bc$$

Trace of A

$$\text{Tr } A = \sum a_{ii}$$

$$= a + d$$

= sum of diagonal entries

$$\text{if } T^2 - 4D \begin{cases} > 0 & \text{real dist.} \\ = 0 & \text{repeated} \\ < 0 & \text{complex.} \end{cases}$$

$$= \lambda^2 - (a+d)\lambda + (ad-bc)$$

Det of A:

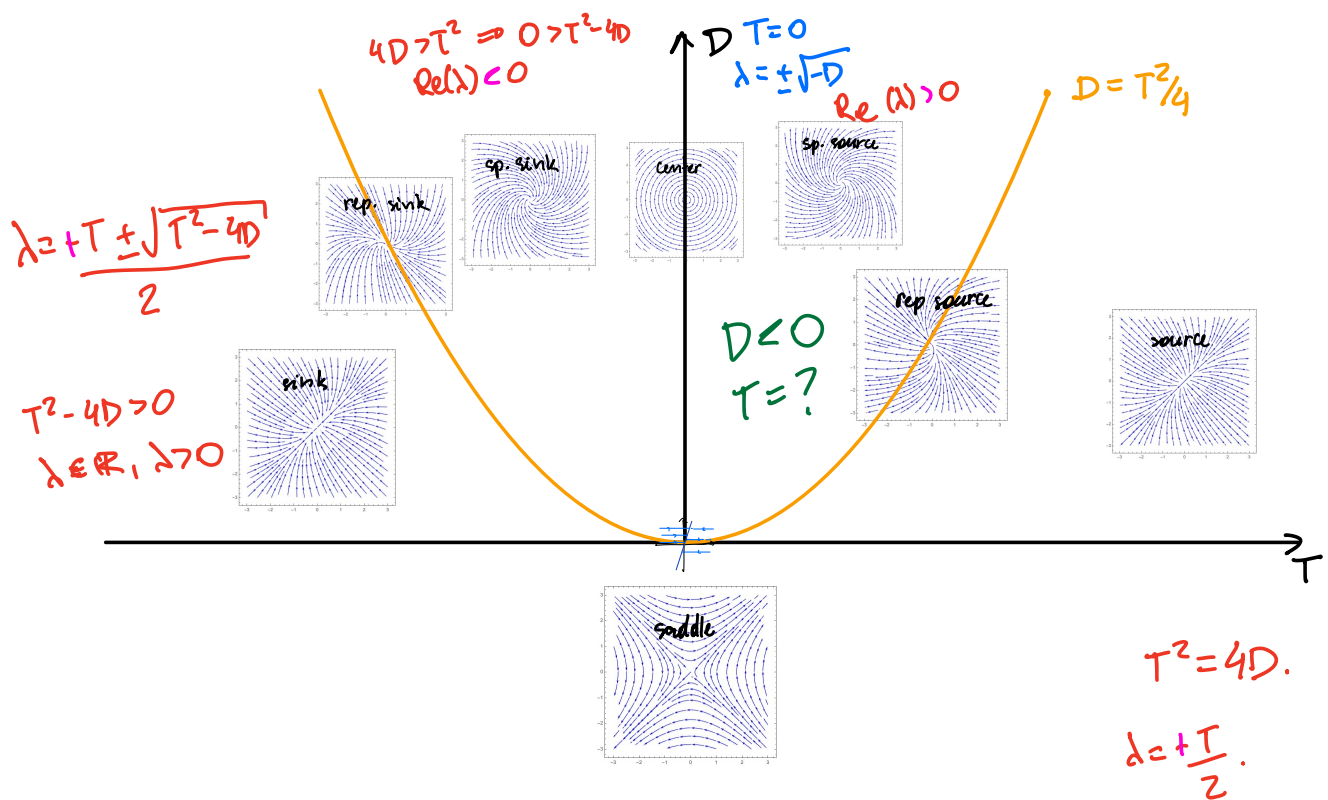
$$\det A = ad - bc$$

$$= \lambda^2 - \text{Tr } A \lambda + \det A$$

$$= \lambda^2 - T\lambda + D$$

$$\lambda = \frac{-T \pm \sqrt{\dots}}{2} \Rightarrow \text{if } T \begin{cases} < 0 \\ = 0 \\ > 0 \end{cases}$$

Extra motivation: I don't want to compute eigenvalues, can we use just that?



$$\det(A) = \det\left(P \cdot \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \cdot P^{-1}\right)$$

$$= \cancel{\det(P)} \cdot \det \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \cdot \cancel{\det(P^{-1})}$$

$$= \lambda_1 \lambda_2$$

if $\det(A) < 0 \Rightarrow \lambda_1 \lambda_2 < 0 \Rightarrow$ so λ_1 and λ_2
 have opposite signs.

Q: How useful it is?

the biggest
limitations:

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{cases} x' = y \\ y' = -x \end{cases}$$

$$\det A = 1$$

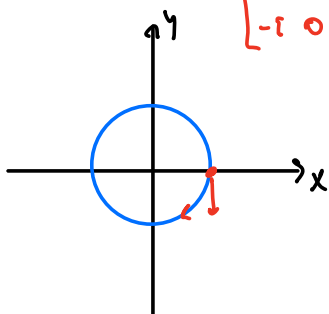
$$\operatorname{tr} A = 0$$

$$\det B = 1$$

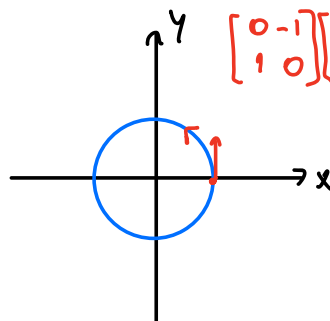
$$\operatorname{tr} B = 0$$

$$\begin{cases} x' = -y \\ y' = x \end{cases}$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

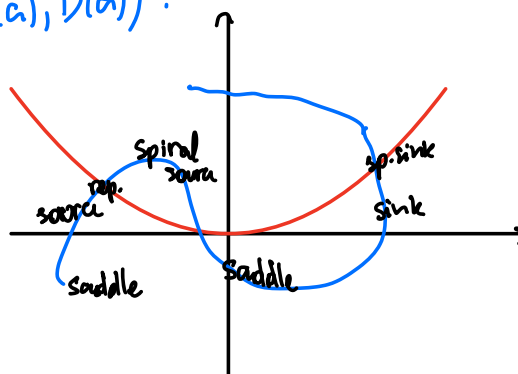


$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



cons: for linear systems with parameter, we can sketch
the curve $(T(a), D(a))$:

$$X' = A(a)X$$



Ex. Consider the one-parameter family of linear sys:

$$x' = -2x + ay$$

$$y' = -2x$$

$$\text{tr} A = -2$$

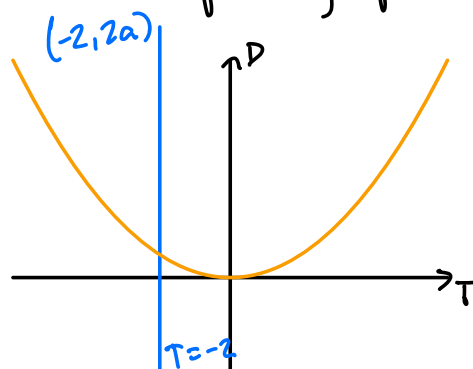
$$\det A = 2a$$

What are the bifurcations points of this system?

$a=0$ since you are crossing $D=0$.

$$T^2 - 4D = 0 \Rightarrow 4 - 4 \cdot 2a = 0$$

$$a = \frac{1}{2}$$



Ex. Consider the one-parameter family of linear sys:

$$x' = ax - y$$

$$y' = y + ay$$

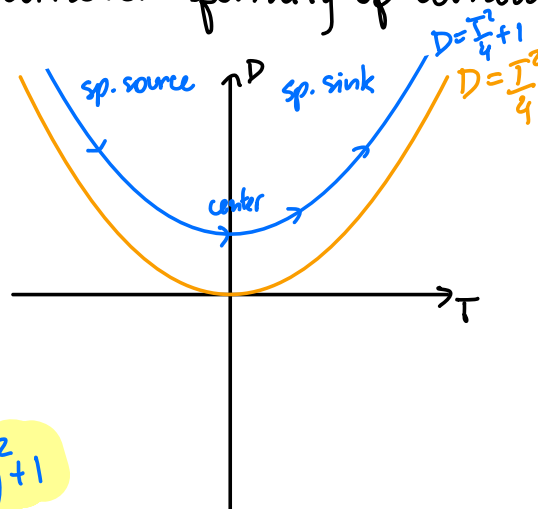
$$A = \begin{bmatrix} a & -1 \\ 1 & a \end{bmatrix}$$

$$\det A = a^2 + 1$$

$$\text{tr } A = 2a$$

$$a = \frac{I}{2}$$

$$D = \left(\frac{I}{2}\right)^2 + 1$$



Bernardo's claim: there is ALWAYS a single equilibrium point.

$$AX = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \det A = a^2 + 1 > 1 > 0$$

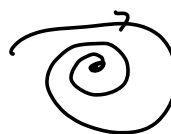
spiral source



center



sp. sink



claim: going through the bifurcation there is

birth or death of a periodic solution.

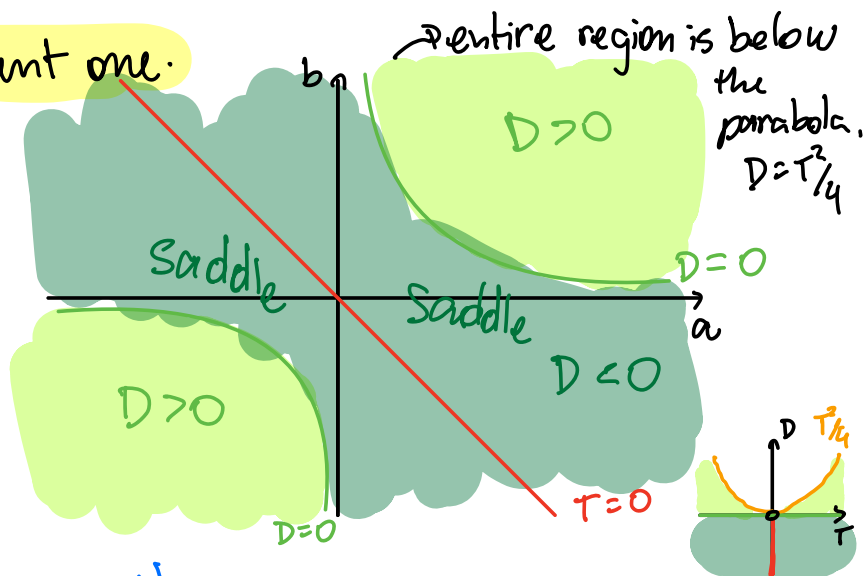
Andronov-Hopf bifurcation.

This (T, D) plane is called "parameter plane".

Example of a different one.

$$\begin{cases} x' = ax + y \\ y' = by + x \end{cases}$$

$$T^2 - 4D = (0+0) - 4(0-1) = 4 > 0$$



$$A = \begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} \Rightarrow \begin{aligned} \text{tr} A &= a+b \\ \det A &= ab-1 \end{aligned}$$

$$\frac{(a+b)^2}{4} = ab-1 \Rightarrow \frac{a^2+2ab+b^2}{4} = ab-1$$

$$D=0 \text{ when } ab=1 \\ b=1/a$$

$$\text{tr} A = 0 \quad \begin{aligned} a+b &= 0 \\ b &= -a \end{aligned}$$

$$\frac{a^2-2ab+b^2}{4} = -1 \Rightarrow (a-b)^2 = -4 \text{ never happens.}$$

end of 3.7. Probably not doing 3.8., so this is the end of Ch. 3.

Feel free to read 3.8 and ask questions though. ;)

Monday: Forced Harmonic Oscillator

$$mx'' + bx' + kx = \underline{F(t)} \neq 0.$$